MIL-M-45625A(Ord)
22 April 1960

SUPERSEDING
MIL-M-45625(Ord)
27 May 1959

MILITARY SPECIFICATION

MIXER STAGE, SWEEP CONTROL: CV-318/M

1. SCOPE

1.1 This specification covers one type of sweep generator, designated as Mixer Stage, Sweep Control: CV-318/M hereinafter referred to as "generator". It is a component of the tracking console used in antiaircraft fire control and guided missile systems (see 6.1.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Ordnance Corps

D7614459	Mixer Stage, Sweep Control: CV-318/M, Schematic Diagram
E7620521	Oscillator, Audio Frequency, 0-296/M
E7620604	Mixer Stage, Sweep Control: CV-318/M

FSC 1285

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one generator representative of production, shall be made available to the Government to determine if the generator meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The generator shall be manufactured in accordance with Drawing E7620604, "Mixer Stage, Sweep Control: CV-318/M" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential.- The design and construction of the generator shall insure that all external parts are at ground potential at all times.

3.4 General specification .- The generator shall meet the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron Tubes
(f) Workmanship

3.5 Environmental.

3.5.1 Vibration.- The generator shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the generator shall not be damaged or adversely affected and shall meet all requirements of this specification

3.5.2 Operating temperatures.- The generator shall meet all the requirements of this specification while thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

3.5.3 Storage temperatures.- The generator shall show no evidence of physical failure and shall meet all performance requirements of this specification at the standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

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3.6 Electrical.

3.6.1 Circuitry.- The circuitry of the generator shall be in accordance with Electrical Schematic Drawing D7614459.

3.6.2 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the generator are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 5 seconds.

3.6.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.7 Performance.- The generator, when set up in accordance with pretest information specified in Section 4, shall meet the following requirements. All symbols referred to in the following paragraphs are those shown on Drawing D7614459.

3.7.1 Static conditions.- Under static conditions, BAL ADJ, R17 shall be capable of differing the output currents from one another by 3 milliamperes. R17 shall also be capable of equalizing the output currents within the range of 11 to 21 milliamperes (ma).

3.7.2 Dynamic operation.

3.7.2.1 Linearity.

3.7.2.1.1 When a 5-volt, 4 kilocycle (kc) signal input (same phase as carrier) is applied, the current difference shall be 27 plus or minus 4 milliampere (ma) (R2 inserted fully into the circuit).

3.7.2.1.2 When a 5-volt, 4-kc signal input (180 degrees out-of-phase with the carrier) is applied, the current difference shall be 27 plus or minus 4 ma.

3.7.2.1.3 When inputs as specified in 3.7.2.1.1 and 3.7.2.1.2 are applied and R2 shorted out of the circuit, the current difference shall be 18 ma (maximum).

3.7.2.2 Limiting.

3.7.2.2.1 The output current at terminal 11 of P1 shall be 45 ma (max), when a 15-volt, 4-kc signal (same phase as carrier) is applied to the generator (R2 fully inserted into circuit).

3.7.2.2.2 The output at terminal 9 of P1 shall be 45 ma (max), when a 15 volt, 4-kc signal (out-of-phase with carrier) is applied to the generator.

3.8 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.4 Production model.

4.4.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

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4.4.2 These tests shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment designed or procured for use on the remaining items.

4.4.3 The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

4.4.4 Disposition of rejected production model.- Should the production model fail to pass the test for Government acceptance, the contractor shall cease those operations which are causing rejectable material and make the necessary corrections. When corrections have been made, a new production model shall be tested by the contractor under the surveillance of the Government. Operations causing rejections shall not be resumed until a production model has successfully passed all inspections specified.

4.5 Inspection provisions.

4.5.1 Inspection of components.- Contractor's inspection for all components shall be in accordance with the inspection procedures of the written inspection plan required by 4.2 and 4.2.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements that have a bearing on the functional accuracy and performance of the unit.

4.5.2 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.3 Disposition of rejected lots.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection of the characteristics that caused the lot to be rejected.

4.6 Inspection testing.

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4.6.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.6.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.6.3 Acceptance tests.

4.6.3.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

4.6.3.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L5, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 4.0%	Reqt Par.	QA Par.	Test Method
1. Physical failure (post vibration)	3.5.1	4.8.1.1	Tactile-Visual
2. Static conditions	3.7.1	4.8.3.1	Standard Meas. Equip. (SME)
3. Linearity	3.7.2.1	4.8.3.2	SME
4. Limiting	3.7.2.2	4.8.3.3	SME

NOTE: The above tests shall be performed while thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit, except the vibration test which will be at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.5%	Reqt Par.	QA Par.	Test Method
1. Circuitry	3.6.1	4.8.2.1	Standard Meas. Equip. (SME)
2. Dielectric strength	3.6.2	4.8.2.2	SME

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TABLE III - CLASSIFICATION OF DEFECTS (CONT'D)

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.5%	Reqt Par.	QA Par.	Test Method
3. Insulation resistance	3.6.3	4.8.2.3	SME
4. Static conditions	3.7.1	4.8.3.1	SME
5. Linearity	3.7.2.1	4.8.3.2	SME
6. Limiting	3.7.2.2	4.8.3.3	SME
7. Identification marking	3.8	4.8.5	Visual

NOTE: The above tests shall be performed at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit)

4.7 Inspection equipment.

4.7.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.7.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.1.1.1 Test equipment accuracy requirements.- In addition to the accuracy requirements specified in 4.7.1.1, the following test equipment shall be connected as shown on Figure 1 and Drawing D7614459, and as specified below:

Item Number	Description
1.	Generator, sine wave, 4 kilocycles plus or minus 10 percent, 3 outputs to work into a 5000 ohm load. (a) 5 volts reversible phase (b) 15 volts reversible phase (c) 70 volts fixed phase Oscillator Audio Frequency O-296/M Ordnance Corps. Drawing E7620521 may be used.

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Item Number (Cont'd)	Description
2.	Voltmeter (V1) 0-90 volts plus or minus 2 percent, frequency range 0 to 150,000 cycles per second.
3.	Ammeter (A1, A2) 0-50 milliamperes plus or minus 2 percent quantity 2.
4.	Resistor, matched pair within plus or minus 1 percent of each other, 10 watts, 2500 ohms plus or minus 5 percent, quantity 2.
5.	Power supply (6.3 volts ac, 400 cycles single phase).
6.	Power supply (plus 250 volts, 0.040 ampere)
7.	Power supply (minus 250 volts, 0.001 ampere)

4.7.1.1.2 Device to test for vibration requirement.- The device shall consist of a vibrating machine which will produce a motion in each of the 3 mutually perpendicular planes at a uniformly varied frequency between the limits of 10 and 55 cycles per second and return to 10 cycles. It shall have an amplitude of 0.03 inch (0.06 inch maximum excursion) and shall be equipped with a medium capable of holding the unit securely to the vibrating machine.

4.7.2 Government provisions.- Test equipment listed in 4.7.1.1.1 will be used for Government product inspection (reference 4.3(b)), and to settle disputes (reference 4.2.1).

4.8 Test methods and procedures.

4.8.1 Environmental.

4.8.1.1 Vibration.- Each generator in the sample lot shall be mounted in its normal operating position on a special testing device and vibrated in accordance with Method 201 of Standard MIL-STD-202. Subsequent to vibration, the generator shall be subjected to a visual and tactile examination for physical damage. The generator shall then be subjected to all the tests specified herein.

4.8.1.2 Operating temperatures.- The test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" and not raised beyond condition "G". When the generator is thermally stabilized at condition "C" and "G", it shall be subjected to the tests specified in 4.8.3.1 to 4.8.3.3 inclusive, and shall meet the requirements specified.

4.8.1.3 Storage temperatures.- This test shall be conducted in accordance with procedure II of Specification MIL-F-13926 to the conditions specified. Upon return to the standard ambient temperature, the generator shall be inspected for damage and then subjected to the tests specified in 4.8.3.1 to 4.8.3.3 inclusive, and shall meet the requirements specified. This test shall be conducted on the production model only.

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4.8.2 Electrical.

4.8.2.1 Circuitry.- A circuitry check of the generator circuits shall be made using standard measuring equipment to determine conformance with the requirements specified.

4.8.2.2 Dielectric strength.- The generator shall be tested in accordance with Method 301 of Standard MIL-STD-202, and shall meet the requirements specified.

4.8.2.3 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the generator shall be given the insulation resistance test in accordance with Method 302, test condition "B", of Standard MIL-STD-202, and shall meet the requirements specified.

4.8.3 Performance.

4.8.3.1 Static conditions.- The AZM. ADJ. control R2 shall be advanced to its full clockwise position.

4.8.3.1.1 The 70 volt, 4 kilocycle carrier from item 1 shall be applied to terminals 1 and 3 of P1.

4.8.3.1.2 With no signal input, the BAL, ADJ. potentiometer R17 shall be rotated from one extreme to the other to make either output current conform to the requirements when measured with item 3. R17 shall be set so that both currents are equal and then locked. The currents shall be within the range as specified.

4.8.3.2 Linearity.

4.8.3.2.1 When the specified signal input is applied, the difference in output currents when measured by ammeters A1 and A2 shall be as specified.

4.8.3.2.2. This procedure shall be repeated when a reversed phase signal input is applied.

4.8.3.2.3 With R2 turned to maximum counterclockwise position, the current difference shall not exceed the maximum as specified.

4.8.3.3 Limiting.- R2 shall be turned to a full clockwise position.

4.8.3.3.1 With the input signal applied as specified, the output current as indicated on ammeter A1 shall not exceed the maximum as specified.

4.8.3.3.2 With the reverse phase signal input applied as specified, the output current as indicated on ammeter A2 shall not exceed the maximum as specified.

4.8.4 Final acceptance.- Final acceptance of a lot of generators shall be withheld by the Government inspector until the sample units, representative of that production lot, have satisfactorily passed the applicable requirement of this specification.

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4.8.5 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

4.8.6 Packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The generator shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 P4136. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The purpose of the generator is to generate an azimuth sweep which occurs during a 30 degree sector of acquisition antenna rotation. The input is a 4000 cycles per second resolver signal.

6.1.1 Application.- The generator is a component of the following assemblies and systems:

Assembly	System
Tracking Console	Antiaircraft Fire Control Systems M33C (90MM) and M33D (120MM)
Tracking Station Group OA-866/MPA-4	Radar Course Directing Control AN/MSW-1 and AN/MSW-3
Tracking Station Group OA-1595/MPA-5	Antiaircraft Guided Missile System - Nike Hercules
Tracking Console OA-606/MSG	Guided Missile Fire Control System AN/MSG-3 (Terrior)

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Definitions.

6.3.1 Standard measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

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6.3.2 Tactile.- Skillful manipulation or sensing.

6.3.3 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

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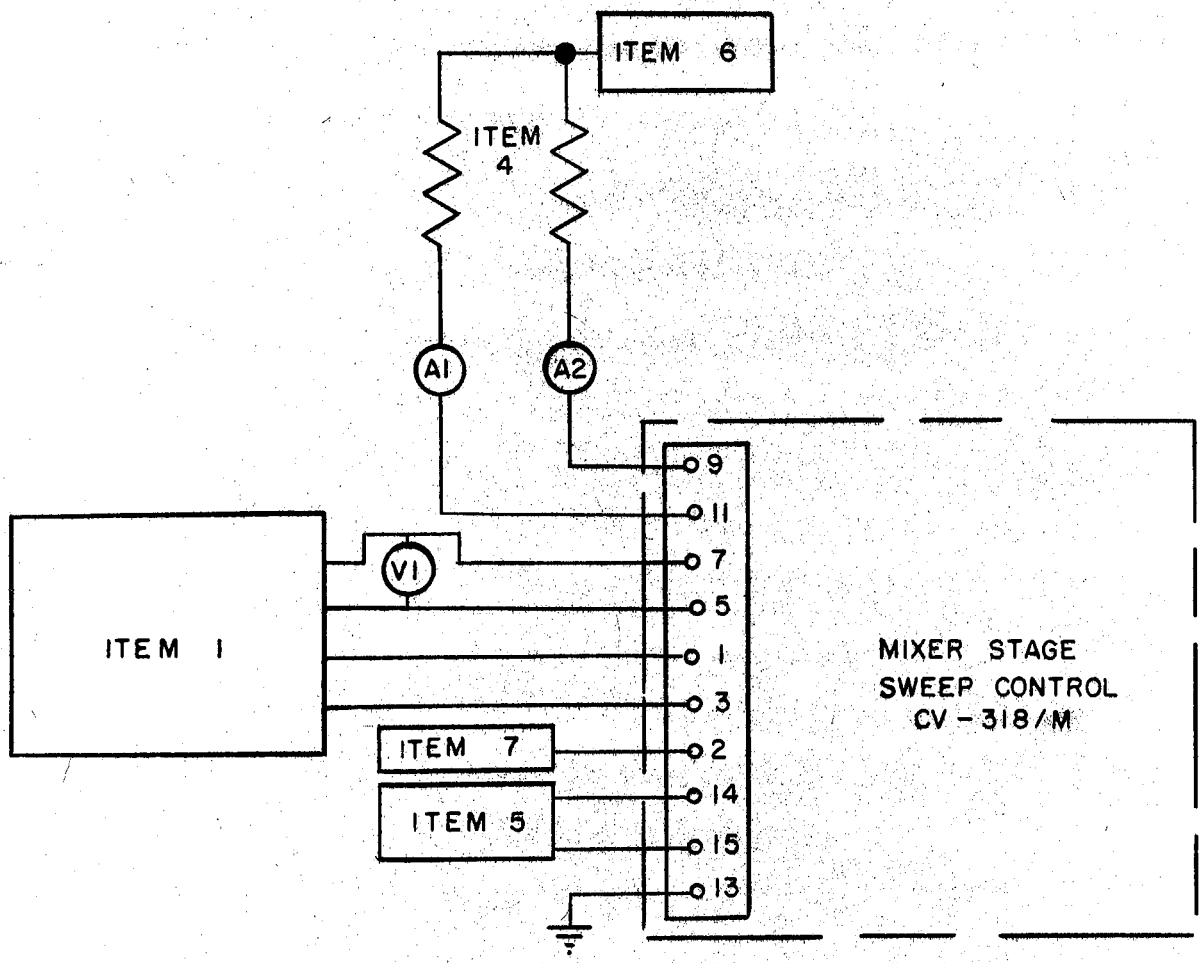
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NOTE:

IN CONNECTING ITEM 1 TO P1, TERMINALS NO 1 AND 5 ARE TO BE THE GROUND SIDE

TEST SET-UP

FIGURE 1

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